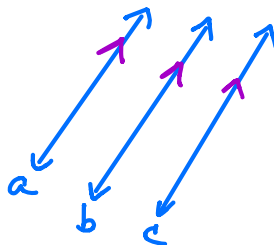


Transitive Property of Parallel Lines Theorem

If $a \parallel b$ and $b \parallel c$,
then $a \parallel c$.



If 2 lines are
parallel to the same
line, then they are
parallel to each other.

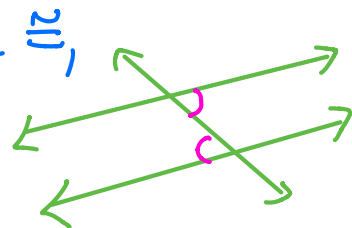
Converse Theorems

★
TO
PROVE
LINES
ARE

PARALLEL

① Alternate Interior Angles CONVERSE Theorem

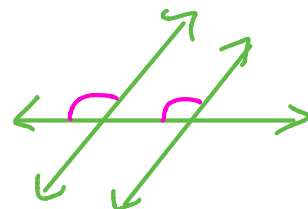
If alternate interior angles are $\cong$,
then the lines are parallel.

**② Alternate Exterior Angles CONVERSE Theorem**

If alternate exterior angles are $\cong$,
then the lines are parallel.

③ Corresponding Angles CONVERSE Theorem

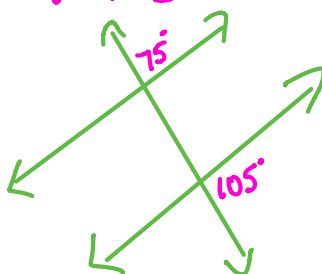
If corresponding angles are $\cong$,
then the lines are parallel.

**④ Consecutive Interior Angles (same-side interior) CONVERSE Theorem**

If consecutive interior angles are supplementary,
then the lines are parallel.

⑤ Consecutive Exterior Angles (same-side exterior) CONVERSE Theorem

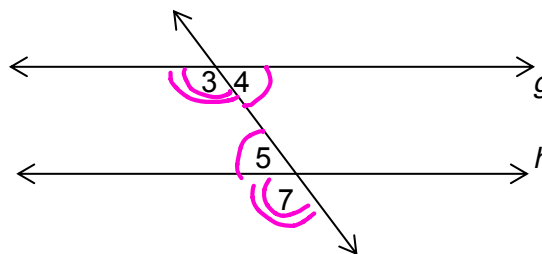
If consecutive exterior angles are supplementary,
then the lines are parallel.



Prove the Alternate Interior Angles Converse Theorem

Given: $\angle 4 \cong \angle 5$

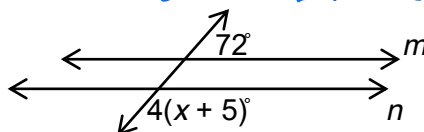
Prove: $g \parallel h$



Statements	Reasons
① $\angle 4 \cong \angle 5$	① given
② $\angle 3$ and $\angle 4$ are supplementary $\angle$'s	② Linear Pair Postulate
③ $\angle 5$ and $\angle 7$ are supplementary $\angle$'s	③ Linear Pair Postulate
④ $\angle 4$ and $\angle 7$ are supp. $\angle$'s	④ Substitution Prop.
⑤ $\angle 3 \cong \angle 7$	⑤ Congruent Supplements Thm
⑥ $g \parallel h$	⑥ Corresponding Angle CONVERSE Thm.

Example Find the value of x that makes $m \parallel n$. State the Theorem.

Consecutive
Exterior Angles
Converse
Theorem



$$4(x+5) + 72 = 180$$

$$4x + 20 + 72 = 180$$

$$x = 22$$

